SMART RESTAURANT MANAGEMENT SYSTEM

Roadmap, Testing and AI Phase Plan

Timeline, team, quality, risks and Phase 2 rollout

DOCUMENT ID	SRMS-10
VERSION	1.0
DATE	06 November 2026
STATUS	Draft for review
PHASE	Phase 1: event-driven core | Phase 2: AI camera detection
PURPOSE	How and when we will deliver, how we will test it, what could go wrong, and how AI camera detection is introduced safely.

1. Delivery Roadmap

Estimates assume a small team (see section 2) and are indicative; they will be refined after design sign-off.

Phase	Weeks	Deliverables	Exit criteria
0. Discovery and design	1 to 2	Approved documents, UI wireframes and design system, API contract, test restaurant details	Sign-off on SRS and UX
1A. Foundation	3 to 5	Auth and roles, restaurants/floors/tables, QR generation, menu admin, CI/CD, staging	Staff can log in and set up a restaurant
1B. Ordering	6 to 9	QR session, menu, shared cart, orders, kitchen display, real-time events	Guest order reaches kitchen in under 2 seconds
1C. Payments and billing	10 to 12	Bill, GST, UPI/card via gateway, cash, receipts, refunds	Payments reconcile in test and live sandbox
1D. Floor control	13 to 16	State engine, live floor map, alerts engine, cleaning queue, basic reports	Full table lifecycle works end to end
1E. Hardening and pilot	17 to 20	Load, security and usability testing, fixes, pilot in one restaurant	Pilot KPIs met; go-live decision
1.5 Enhancements	21 to 24	Waitlist, reservations, feedback, split bill, multi-language	Adoption by pilot staff
2. AI camera detection	25 to 36	Camera zones, detection service, suggestions, auto-update, queue and dwell analytics	Accuracy targets met in pilot
3. Scale	37+	Multi-outlet, loyalty, advanced analytics, native apps if needed	Business decision

2. Suggested Team

Role	Count	Notes
Product owner	1	You; decisions on scope and priorities
UI/UX designer	1	Part-time after Phase 0
Full-stack developers (Node, React)	2 to 3	Core build
QA engineer	1	Automation and manual testing
DevOps (part-time)	1	CI/CD, hosting, monitoring
ML engineer (Phase 2)	1	Vision model, edge deployment

3. Testing Strategy

Level	Scope	Tools
Unit tests	State engine transitions, price and tax calculation, alert rules, permissions	Jest
Integration tests	API with real database: session, order, payment, webhook flows	Jest + Supertest, test containers
End-to-end tests	Guest scan to pay; manager sees states; cleaner resets table	Playwright (mobile and desktop viewports)
Concurrency tests	Two scans at once, duplicate webhooks, payment plus override race	Custom scripts, k6
Load tests	500 concurrent guest sessions and 100 staff sockets per outlet	k6
Security tests	OWASP Top 10 checks, QR token guessing, role bypass, webhook forgery	ZAP, manual review
Usability tests	Real waiters, kitchen and cleaners in pilot; measure taps and errors	Observation and surveys

Level	Scope	Tools
Device tests	Low-end Android phones, iPhone Safari, kitchen tablet	BrowserStack plus real devices

Critical test scenarios

- Table state follows the full cycle Available, Occupied, Payment Pending, Needs Cleaning, Cleaning, Available with correct log entries.
- Payment webhook arrives twice: only one success and one cleaning task.
- Guest closes the browser after paying: table still moves correctly.
- Network loss on the manager screen: after reconnect the map matches the server.
- Unacknowledged alert escalates at the configured time.
- Role tests: cleaner cannot see payments; waiter cannot refund.

4. Phase 2 Plan: AI Camera Detection

4.1 What we will detect

Capability	Output event	Use
People per table zone	PERSON_SEATED / PERSON_LEFT	Auto-occupy and auto-release suggestions; guest count
Table cleared	TABLE_CLEARED	Suggest Needs Cleaning to Available when plates are removed and the cleaner has wiped
Staff near table	STAFF_VISIT	Reset unattended timers
Guest waiting for staff	LONG_UNATTENDED	Raise Needs Attention
Entrance crowd	QUEUE_COUNT	Estimate wait time and feed waitlist
Dwell time	Computed from events	Analytics and long-stay alerts

4.2 Rollout steps

1. Install or reuse existing CCTV; verify camera angles cover tables without privacy-sensitive areas.
2. Calibrate: draw table zones on a still frame for each camera.
3. **Shadow mode:** the AI only logs events, and we compare to real table status for 2 to 4 weeks.
4. **Suggestion mode:** staff see suggestions and confirm or dismiss; measure precision and recall.
5. **Auto mode:** high-confidence events update status automatically; manual override always wins.
6. Continuous tuning using confirmed and dismissed suggestions.

4.3 Targets and safeguards

- Targets for auto mode: at least 95% precision on occupied/free detection; clear fallback when cameras are offline.
- Privacy: no face recognition; process on the edge; store only counts and states; snapshots optional with short retention; visible notice to guests and staff as required by law and policy.
- Reliability: heartbeats from each camera; if a camera is offline, tables in its zone fall back to event-driven logic automatically.

5. Risks and Mitigations

Risk	Impact	Mitigation
Staff forget to tap Cleaned or Served	Wrong map status	Timers and reminders; gamified cleaning times; Phase 2 AI confirms
Guests do not scan or prefer waiter ordering	Lower adoption	Waiter order screen; table tent cards; small incentive; simple UI

Risk	Impact	Mitigation
Poor restaurant internet	Missed updates	Resync on reconnect; offline banners; backup mobile hotspot recommendation
Payment edge cases (webhook delay, duplicates)	Financial errors	Idempotency, webhook-first confirmation, daily reconciliation
QR misuse (photo of QR, remote ordering)	Fake orders	Session binding, rate limits, optional waiter confirmation, location hint or short code on table
Scope creep	Delay	Strict phase gates; backlog prioritisation by product owner
Camera accuracy and privacy concerns (Phase 2)	Wrong states, legal risk	Shadow mode, confidence thresholds, no biometric data, clear signage
Staff resistance to change	Poor usage	Training, pilot champions, simple one-tap actions

6. Open Decisions

Decision	Options	Recommendation
Payment gateway	Razorpay, Cashfree, PayU	Razorpay for UPI-first and docs; confirm fees
Guest identity	Anonymous QR only, or phone OTP	Anonymous first; optional phone for receipt and waitlist
Waiter confirmation of QR orders	Auto-send to kitchen, or waiter confirms	Configurable; default auto-send
Language support at launch	English only, or plus Hindi and Gujarati	English at launch, add Hindi and Gujarati in 1.5
Hosting region	India region of chosen cloud	India region for latency and data residency
Number of outlets at launch	One pilot outlet	One, with multi-tenant schema from day one

7. Next Steps

1. Review and approve this document set (overview, roles, modules, schema, SRS, workflows, architecture, API, UI/UX).
2. Confirm open decisions in section 6.
3. Produce clickable UI prototype for the guest app and the manager floor map.
4. Set up repository, CI/CD and staging, then start Phase 1A.